

Wasmir Chowdhury

wchowdhu@andrew.cmu.edu | [linkedin.com/in/wc0409](https://www.linkedin.com/in/wc0409) | (646) 578-1035

EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Integrated BS/MS in Electrical and Computer Engineering, GPA: 3.21

May 2028

Relevant Coursework

Embedded Systems, Real-Time Operating Systems, Structure and Design of Digital Systems, Computer Systems

WORK EXPERIENCE

Undergraduate Research Assistant

May 2026 – Present

WiseLAB (Carnegie Mellon University)

Pittsburgh, PA

- Setting up two ZED X One GS cameras as a virtual stereo pair on a Jetson AGX Orin robotics compute platform, debugging Linux kernel sensor enumeration issues to bring up ZED's hardware-synced (15 μ s) capture stack
- Building a real-time stereo H.265 capture pipeline in GStreamer using Jetson hardware encoder nodes, encoding both camera streams on-device at record time with no intermediate transcode

Course Teaching Assistant

August 2026 – Present

18-349: Introduction to Embedded Systems (Carnegie Mellon University)

Pittsburgh, PA

- Lead two weekly recitations (5 students each) on embedded architectures and real-time scheduling
- Hold 5 hours of office hours weekly, guiding students through ARM Cortex-M labs on interrupts, memory management, and RTOS development

PROJECT EXPERIENCE

Real-Time Operating System (RTOS) Kernel | C, ARM Assembly

- Built a preemptive RMS scheduler on ARM Cortex-M4 with UB-test-based admission control
- Implemented HLP/IPCP mutex locking in C to prevent priority inversion and deadlocks across concurrent tasks
- Wrote a PendSV context switcher in ARM Assembly to manage thread state and CPU register save and restore

QNX Real-Time Vibration Monitor (Hackathon) | Python, QNX RTOS

3rd Place, QNX Track

- Built a real-time vibration monitor on Raspberry Pi 5 with QNX RTOS, sampling an MPU6050 via I2C at 100Hz
- Navigated QNX hardware bring-up including I2C resource manager limits, MPU6050 wake sequencing, and GPIO drivers, using slog2 logging and POSIX signals

STM32 Bare-Metal Peripheral Drivers | C, STM32, ARM Cortex-M

- Implemented interrupt-driven I2C and UART drivers from scratch via direct peripheral register configuration
- Debugged hardware interface timing using GDB over JTAG and an oscilloscope to validate driver correctness

Malloc Lab | C

- Implemented a custom dynamic memory allocator in C, replacing libc's malloc, free, and realloc
- Iteratively optimized from an implicit free list to segregated free lists with coalescing, footerless blocks, and mini blocks

Matrix Multiply Accelerator (18-341) | SystemVerilog, Vivado, FPGA

- Designed a pipelined MAC accelerator in SystemVerilog on a Spartan-7 FPGA using multiplier IP and Block RAM
- Met a 100MHz timing constraint via Vivado STA, finishing in under 525 clock cycles (512-cycle target)

LEADERSHIP EXPERIENCE

EMERGE

July 2024 – May 2026

Mentor, Carnegie Mellon University

Pittsburgh, PA

- Facilitate monthly group meetings with mentees to offer academic, personal, and professional guidance

TECHNICAL SKILLS

Embedded Systems: STM32, ARM Cortex-M, Computer Architecture, RTOS (RMS, HLP/IPCP), Bare-Metal Programming, Memory Management, I2C, UART, Interrupts, GDB, JTAG, Oscilloscope, Logic Analyzer

Software: C, C++, ARM Assembly, Python, Shell/Bash, SystemVerilog, FPGA, Linux, Data Structures, Git/GitHub, Vivado, Fusion360 PCB